

Algebraic Geometry and Coding Theory

Lecture 2

September 17

Main theme. After completing the proof of the asymptotic Plotkin bound, we begin the geometric part of the course. Affine and projective curves have the same rational functions, but projective curves allow us to account for all their zeros and poles. Over a finite field, Frobenius orbits organize points of higher degree. Divisors record these points with multiplicities, and Riemann–Roch controls spaces of rational functions with prescribed zeros and poles.

We write $\bar{k}$ for an algebraic closure of a field k , and $\bar{C}$ for the projective closure of an affine plane curve C . These are different uses of the bar. The letter q denotes a prime power.

Contents

1	Completing the asymptotic Plotkin bound	2
2	Fields, extensions, and algebraic closures	3
2.1	Algebraic extensions and algebraically closed fields	3
2.2	Finite fields and the algebraic closure of a finite field	4
3	Affine and projective plane curves	5
3.1	Affine curves and their points	5
3.2	Projective space and homogeneous equations	6
3.3	Homogenization and projective closure	6
4	Smooth and singular points	7
5	Coordinate rings and function fields	8
5.1	The function field of an affine curve	8
5.2	The function field of a projective curve	9
5.3	Passing to projective closure does not change the function field	10
6	Bézout’s theorem: counting all intersections	11
7	Frobenius and points of higher degree	12
7.1	Frobenius orbits	12
7.2	A cubic over $\mathbb{F}_3$	13
8	From intersection points to divisors	14
9	Divisors of rational functions	16
9.1	Zeros minus poles	16
9.2	Explicit principal divisors	17
10	Riemann–Roch spaces	18
10.1	Rational functions associated to a divisor	18
10.2	The Riemann–Roch theorem	19
10.3	Riemann–Roch on the running cubic	20

1 Completing the asymptotic Plotkin bound

Recall from Lecture 1 that $A_q(n, d)$ is the largest size of a code in $\mathbb{F}_q^n$ whose minimum distance is at least d . We allow the convention $A_q(n, 0) = q^n$. For $0 \leq \delta \leq 1$, set

$$\alpha_q(\delta) = \limsup_{n \rightarrow \infty} \frac{1}{n} \log_q A_q(n, \lfloor \delta n \rfloor), \quad \theta = 1 - \frac{1}{q}.$$

The finite Plotkin bound proved last time says

$$A_q(n, d) \leq \frac{d}{d - \theta n} \quad \text{if } d > \theta n. \quad (1)$$

The asymptotic version was stated without proof. We now prove it by the shortening argument in Walker, §2.2, keeping track of the integer parts and the endpoint $\delta = \theta$.

Lemma 1.1 (Shortening). *Let $\mathcal{C} \subseteq \mathbb{F}_q^n$ have M codewords, with every two distinct codewords at distance at least d . For each integer m with $0 \leq m \leq n$, there is a code $\mathcal{C}' \subseteq \mathbb{F}_q^m$ with at least M/q^{n-m} codewords and with every two distinct codewords still at distance at least d .*

Proof. There are q^{n-m} possible patterns in the last $n - m$ coordinates. By the pigeonhole principle, some pattern occurs in at least M/q^{n-m} codewords. Keep those codewords and delete their last $n - m$ coordinates.

The deleted coordinates agree for every pair of retained codewords, so deleting them does not change any of their pairwise distances. In particular, distinct retained codewords remain distinct. $\square$

Remark 1.2. Shortening is not the same as deleting coordinates from every codeword. If we simply delete coordinates from the whole code, its minimum distance can decrease. In shortening, we first restrict to codewords that agree on the coordinates to be deleted; this is why the distance is preserved.

Theorem 1.3 (Asymptotic Plotkin bound). *For $\theta = 1 - 1/q$,*

$$\alpha_q(\delta) \leq 1 - \frac{\delta}{\theta} \quad (0 \leq \delta \leq \theta),$$

and

$$\alpha_q(\delta) = 0 \quad (\theta \leq \delta \leq 1).$$

Proof. First suppose $\delta > \theta$. Put $d_n = \lfloor \delta n \rfloor$. For all sufficiently large n ,

$$d_n - \theta n \geq (\delta - \theta)n - 1 > 0.$$

By (1),

$$A_q(n, d_n) \leq \frac{d_n}{d_n - \theta n} \leq \frac{\delta n}{(\delta - \theta)n - 1}.$$

The right-hand side is bounded as $n \rightarrow \infty$. Taking logarithms and dividing by n gives $\alpha_q(\delta) = 0$.

Now let $0 < \delta \leq \theta$. Choose a code $\mathcal{C}_n$ of size

$$M_n = A_q(n, d_n), \quad d_n = \lfloor \delta n \rfloor,$$

and set

$$m_n = \left\lfloor \frac{d_n - 1}{\theta} \right\rfloor.$$

For all sufficiently large n , we have $1 \leq m_n \leq n$. Moreover,

$$\theta m_n \leq d_n - 1, \quad d_n - \theta m_n \geq 1. \quad (2)$$

Shortening to length m_n produces a code with M'_n words, where

$$M'_n \geq \frac{M_n}{q^{n-m_n}},$$

and all distinct pairs remain at distance at least d_n .

If $M'_n = 1$, then certainly $M'_n \leq n$. Otherwise the finite Plotkin argument applies to the shortened code, using d_n as a lower bound for its distances. By (2),

$$M'_n \leq \frac{d_n}{d_n - \theta m_n} \leq d_n \leq n.$$

Thus in either case

$$M_n \leq nq^{n-m_n}, \quad \frac{1}{n} \log_q M_n \leq 1 - \frac{m_n}{n} + \frac{\log_q n}{n}.$$

Since $d_n/n \rightarrow \delta$, the definition of m_n gives

$$\frac{m_n}{n} \rightarrow \frac{\delta}{\theta}.$$

Taking the limsup proves $\alpha_q(\delta) \leq 1 - \delta/\theta$. At $\delta = \theta$, this gives $\alpha_q(\theta) \leq 0$, hence equality because α_q is nonnegative. Finally, $\alpha_q(0) = 1$ directly from $A_q(n, 0) = q^n$. $\square$

The main step is to make the shortened length small enough that its relative distance lies *above* the finite Plotkin threshold. The distance lower bound stays fixed while the length decreases.

We now turn to geometry. Reed–Solomon codes arose from a space of polynomials whose dimension and number of zeros were controlled by degree. Our next goal is to understand analogous spaces of functions on algebraic curves.

2 Fields, extensions, and algebraic closures

2.1 Algebraic extensions and algebraically closed fields

Definition 2.1. A *field* is a commutative ring k with $1 \neq 0$ in which every nonzero element has a multiplicative inverse. A *field extension* K/k means that k is a subfield of K . Its degree is

$$[K : k] = \dim_k K,$$

which may be infinite.

For a prime p , the residue ring $\mathbb{F}_p = \mathbb{Z}/p\mathbb{Z}$ is a field. A finite field has prime characteristic: if its characteristic were a product ab with $1 < a, b < ab$, then $(a \cdot 1)(b \cdot 1) = 0$ would give zero divisors. Thus it contains a copy of $\mathbb{F}_p$.

Definition 2.2. An element $\alpha \in K$ is *algebraic over k* if some nonzero polynomial $h(T) \in k[T]$ satisfies $h(\alpha) = 0$. The extension K/k is *algebraic* if every element of K is algebraic over k .

For example, $\sqrt{2}$ is algebraic over $\mathbb{Q}$ because it satisfies $T^2 - 2 = 0$. Every finite extension is algebraic: if $[K : k] = r$, then $1, \alpha, \dots, \alpha^r$ are linearly dependent over k , giving a polynomial equation for α .

Definition 2.3. A field k is *algebraically closed* if every nonconstant polynomial in $k[T]$ has a root in k . Equivalently, every nonconstant polynomial splits into linear factors over k . An *algebraic closure* of k is an algebraic extension $\bar{k}/k$ such that $\bar{k}$ is algebraically closed.

The equivalence in the definition follows by successively factoring out a linear factor once a root has been found. The requirement that the polynomial be *nonconstant* is essential.

Theorem 2.4 (Existence and uniqueness of algebraic closures). *Every field k has an algebraic closure. Any two algebraic closures of k are isomorphic by an isomorphism fixing k .*

We use this field-theoretic theorem without proof. The isomorphism is not asserted to be unique. This is the algebraic-closure input used in Walker, §3.1, and Morandi, the opening of Chapter 2.

By the fundamental theorem of algebra, $\mathbb{C}$ is algebraically closed. In fact, $\mathbb{C}$ is an algebraic closure of $\mathbb{R}$, since $[\mathbb{C} : \mathbb{R}] = 2$. In contrast, $\mathbb{R}$ is not algebraically closed because $T^2 + 1$ has no real root.

No finite field is algebraically closed. If its elements are $a_1, \dots, a_N$, the polynomial

$$\prod_{i=1}^N (T - a_i) + 1$$

has value 1 at every element of the field and therefore has no root there.

2.2 Finite fields and the algebraic closure of a finite field

The finite-field facts needed below are those recalled in Walker, Appendix B. We give the root-set construction to explain both existence and the notation $\mathbb{F}_{p^n}$.

Theorem 2.5 (Classification of finite fields). *For every prime p and integer $n \geq 1$, there is a field with exactly p^n elements, unique up to isomorphism. Every finite field has this form.*

Proof. Fix an algebraic closure $\overline{\mathbb{F}_p}$ and consider the roots of

$$h(T) = T^{p^n} - T.$$

Its derivative is $h'(T) = -1$, so it has p^n distinct roots in $\overline{\mathbb{F}_p}$. Let S be this root set.

In characteristic p , $(a + b)^p = a^p + b^p$. Iterating gives the corresponding identity for the p^n -th power. Consequently, if $a, b \in S$, then

$$(a + b)^{p^n} = a + b, \quad (ab)^{p^n} = ab.$$

The set S also contains 0 and 1, is closed under additive inverses, and, for $a \neq 0$, satisfies

$$(a^{-1})^{p^n} = (a^{p^n})^{-1} = a^{-1}.$$

Thus S is a field with exactly p^n elements.

Conversely, a finite field K has some prime characteristic p and is a finite-dimensional vector space over $\mathbb{F}_p$. If its dimension is n , then $|K| = p^n$. Every nonzero $a \in K$ satisfies $a^{p^n-1} = 1$ by Lagrange's theorem for the finite group $K^\times$, so every element satisfies $a^{p^n} = a$. Therefore K is a splitting field of $T^{p^n} - T$ over $\mathbb{F}_p$. The uniqueness theorem for splitting fields gives uniqueness up to isomorphism. $\square$

Remark 2.6. For $n > 1$, the field $\mathbb{F}_{p^n}$ is not the ring $\mathbb{Z}/p^n\mathbb{Z}$. The latter has zero divisors and characteristic p^n , whereas $\mathbb{F}_{p^n}$ has characteristic p .

Inside the fixed algebraic closure, we can take

$$\mathbb{F}_{p^n} = \{a \in \overline{\mathbb{F}_p} : a^{p^n} = a\}.$$

These subfields satisfy

$$\mathbb{F}_{p^a} \subseteq \mathbb{F}_{p^b} \iff a \mid b. \quad (3)$$

Indeed, if $b = ar$, repeatedly applying the p^a -power map proves the inclusion. Conversely, an inclusion makes $\mathbb{F}_{p^b}$ a vector space over $\mathbb{F}_{p^a}$; counting elements gives $p^b = (p^a)^r$ for its dimension r , hence $b = ar$.

Proposition 2.7. *With all finite fields viewed inside a common algebraic closure,*

$$\overline{\mathbb{F}_p} = \bigcup_{n \geq 1} \mathbb{F}_{p^n}.$$

More generally, $\overline{\mathbb{F}_q} = \bigcup_{n \geq 1} \mathbb{F}_{q^n}$.

Proof. Every element $\alpha \in \overline{\mathbb{F}_p}$ is algebraic over $\mathbb{F}_p$, so $\mathbb{F}_p(\alpha)$ is a finite extension and therefore a finite field. It is the unique subfield of $\overline{\mathbb{F}_p}$ of its cardinality, hence equals $\mathbb{F}_{p^n}$ for some n . This proves the first equality. The same argument with $\mathbb{F}_q$ proves the second. $\square$

This union is not a chain indexed by successive integers: $\mathbb{F}_{p^2}$ is not a subfield of $\mathbb{F}_{p^3}$. Instead, any two fields $\mathbb{F}_{p^a}$ and $\mathbb{F}_{p^b}$ lie in a common larger one, for example $\mathbb{F}_{p^{ab}}$. The algebraic closure is infinite, even though each of its elements lies in a finite subfield.

3 Affine and projective plane curves

The geometric definitions in this section follow Walker, Chapter 3, and Morandi, §§2.1–2.2 and §2.5. We distinguish a curve defined over k from its set of k -rational points.

3.1 Affine curves and their points

The *affine plane* over k is $\mathbb{A}^2(k) = k^2$. A nonconstant polynomial $f(x, y) \in k[x, y]$ defines an affine plane curve C . For every extension field K/k , its K -rational points are

$$C(K) = \{(a, b) \in K^2 : f(a, b) = 0\}.$$

The coefficients of the equation lie in k , but its solutions may require a larger field.

Example 3.1. The equation $y - x^2 = 0$ defines the parabola. Its points over K are exactly

$$C(K) = \{(a, a^2) : a \in K\}.$$

The equation $y^2 = x^3 - x$ defines another affine curve; its points require $x^3 - x$ to be a square in the field in question. The equation $y^2 = x^3$ defines the cuspidal cubic. We will distinguish these last two curves using smoothness.

Remark 3.2. Over a finite field, the curve should not be identified with just the finite set $C(\mathbb{F}_q)$. Polynomial identities in its coordinate ring are not determined merely by values on that finite set. For example, $x^q - x$ vanishes at every element of $\mathbb{F}_q$, but is not the zero polynomial. Keeping track of extension-field points avoids this confusion.

A polynomial f is *irreducible over k* if it cannot be written as a product of two nonconstant polynomials in $k[x, y]$. It is *absolutely irreducible* if it remains irreducible in $\bar{k}[x, y]$. These conditions need not agree: $x^2 + y^2$ is irreducible over $\mathbb{R}$ but factors as $(x + iy)(x - iy)$ over $\mathbb{C}$.

When constructing a coordinate ring and its field of fractions, irreducibility over k will suffice. In the divisor and Riemann–Roch discussion, we will work with smooth projective absolutely irreducible curves over a finite field.

3.2 Projective space and homogeneous equations

Definition 3.3. The *projective space* $\mathbb{P}^n(k)$ is

$$\mathbb{P}^n(k) = (k^{n+1} \setminus \{0\}) / \sim,$$

where

$$(a_0, \dots, a_n) \sim (b_0, \dots, b_n) \iff (b_0, \dots, b_n) = \lambda(a_0, \dots, a_n)$$

for some $\lambda \in k^\times$. We write an equivalence class as $[a_0 : \dots : a_n]$.

At least one coordinate must be nonzero. Equivalently, the points of $\mathbb{P}^n(k)$ are the one-dimensional linear subspaces of k^{n+1} .

On the subset where $a_i \neq 0$, divide all coordinates by a_i . The i -th coordinate becomes 1, and the other n coordinates identify this subset with $\mathbb{A}^n(k)$. These subsets are the standard affine charts.

For the projective line,

$$\mathbb{P}^1(k) = \{[a : 1] : a \in k\} \cup \{[1 : 0]\}.$$

The last point is denoted ∞ . In the projective plane, the chart $Z \neq 0$ is the affine plane under

$$(x, y) \mapsto [x : y : 1].$$

The complement $Z = 0$ is a projective line, called the line at infinity for this chart. In particular,

$$\#\mathbb{P}^1(\mathbb{F}_q) = q + 1, \quad \#\mathbb{P}^2(\mathbb{F}_q) = q^2 + q + 1.$$

Definition 3.4. A polynomial $F(X, Y, Z)$ is *homogeneous of degree d* if every monomial with nonzero coefficient has total degree d . A projective plane curve defined over k is given by an equation

$$F(X, Y, Z) = 0$$

for a nonconstant homogeneous polynomial $F \in k[X, Y, Z]$.

Homogeneity gives

$$F(\lambda X, \lambda Y, \lambda Z) = \lambda^d F(X, Y, Z).$$

Thus the condition $F(P) = 0$ is independent of the representative of $P \in \mathbb{P}^2(k)$. The numerical value of F itself is generally *not* independent of that representative.

3.3 Homogenization and projective closure

Let $f(x, y) \in k[x, y]$ have total degree d . Its *homogenization* is

$$F(X, Y, Z) = Z^d f(X/Z, Y/Z). \quad (4)$$

This is a polynomial: a term $a_{ij}x^i y^j$ becomes $a_{ij}X^i Y^j Z^{d-i-j}$. Every resulting term has degree d .

The projective curve $F = 0$ is the *projective closure* $\overline{C}$ of the affine curve $C : f = 0$. Since

$$F(x, y, 1) = f(x, y),$$

the part of $\overline{C}$ in the chart $Z \neq 0$ is exactly C . The remaining points are its points at infinity.

Example 3.5. The equations appearing in this lecture homogenize as follows:

$$\begin{array}{ll} y - x^2 & \longmapsto YZ - X^2, \\ y^2 - x^3 + x & \longmapsto Y^2Z - X^3 + XZ^2, \\ y^2 - x^3 & \longmapsto Y^2Z - X^3, \\ y^2 - x^3 + 3x & \longmapsto Y^2Z - X^3 + 3XZ^2. \end{array}$$

For each of these curves, setting $Z = 0$ forces $X = 0$. There is therefore a unique point at infinity,

$$P_\infty = [0 : 1 : 0].$$

The point $[0 : 0 : 0]$ is never allowed in projective space.

Adding points at infinity is not always the addition of a single point. For comparison, the affine curve $xy = 1$ homogenizes to $XY = Z^2$; at $Z = 0$ it has the two points $[1 : 0 : 0]$ and $[0 : 1 : 0]$.

4 Smooth and singular points

We use formal partial derivatives: the usual polynomial differentiation rules apply over any field, with the integer coefficients interpreted in that field. Thus, for example, the derivative of x^p is zero in characteristic p .

Definition 4.1. A geometric point $P = (a, b) \in C(\bar{k})$ on an affine plane curve $f = 0$ is *singular* if

$$f(P) = f_x(P) = f_y(P) = 0.$$

Otherwise it is *nonsingular*, or *smooth*. The curve is smooth if all its geometric points are smooth.

For a projective plane curve $F = 0$, a point $P \in \mathbb{P}^2(\bar{k})$ is singular if

$$F(P) = F_X(P) = F_Y(P) = F_Z(P) = 0.$$

These are the plane-curve criteria used in Walker, §4.1, and Morandi, §2.4. Smoothness must be checked over the algebraic closure and, for a projective curve, also at infinity. Checking only the rational points over the base field is not sufficient.

Example 4.2 (The parabola). For $f = y - x^2$, one has $f_y = 1$, so the affine parabola is smooth in every characteristic. For its projective closure $F = YZ - X^2$, the partial derivatives are

$$F_X = -2X, \quad F_Y = Z, \quad F_Z = Y.$$

A singular point would have $Y = Z = 0$, and then the curve equation forces $X = 0$. This is not a projective point. Thus the projective parabola is smooth in every characteristic as well.

Example 4.3 (The cuspidal cubic). For $f = y^2 - x^3$,

$$f_x = -3x^2, \quad f_y = 2y.$$

At $(0, 0)$ both derivatives vanish in every characteristic, and the point lies on the curve. Thus the curve is singular. For the projective closure $Y^2Z = X^3$, its point at infinity $P_\infty = [0 : 1 : 0]$ is smooth because $F_Z(P_\infty) = 1$.

Example 4.4 (The curve $y^2 = x^3 - x$). Assume $\text{char}(k) \neq 2$. A singular affine point would have $y = 0$ and

$$x^3 - x = 0, \quad 1 - 3x^2 = 0.$$

The first equation gives $x = 0, 1, -1$. The second expression is 1 at 0 and -2 at ± 1 , so it never vanishes at these points. At infinity, $F_Z(P_\infty) = 1$. Hence the projective curve is smooth, including in characteristic 3.

In characteristic 2, the same equation has a singular point at $(1, 0)$. Thus the characteristic hypothesis matters.

Example 4.5 (Characteristic dependence for $y^2 = x^3 - 3x$). Let

$$f = y^2 - x^3 + 3x, \quad f_x = 3(1 - x^2), \quad f_y = 2y.$$

If $\text{char}(k) \neq 2, 3$, a singular affine point must have $y = 0$ and $x = \pm 1$. But $f(1, 0) = 2$ and $f(-1, 0) = -2$, so neither is on the curve. There are no singular affine points.

In characteristic 3, the equation reduces to $y^2 = x^3$, with a singular point at $(0, 0)$. In characteristic 2, it becomes $y^2 = x^3 + x$, with a singular point at $(1, 0)$.

The homogeneous equation is

$$F = Y^2Z - X^3 + 3XZ^2.$$

Its only point at infinity is P_∞ , and $F_Z(P_\infty) = 1$ in every characteristic. Therefore the projective curve is smooth exactly when the characteristic is different from 2 and 3.

The computations in Example 4.5 also show why a familiar equation over $\mathbb{C}$ cannot simply be declared smooth over every finite field: both the equation and its derivatives change after reduction modulo a prime.

5 Coordinate rings and function fields

5.1 The function field of an affine curve

Let C be defined by an irreducible polynomial $f(x, y) \in k[x, y]$.

Definition 5.1. The *coordinate ring* of C is

$$k[C] = k[x, y]/(f).$$

Since $k[x, y]$ is a unique factorization domain, an irreducible element f is prime. Thus $k[C]$ is an integral domain. Its field of fractions

$$k(C) = \text{Frac}(k[C])$$

is the *function field* of C .

This construction is over the field of definition k . In particular, it uses the relation $f = 0$, not the ideal of all polynomials that happen to vanish on the finite set $C(k)$.

Concretely, a rational function is represented by g/h with $g, h \in k[x, y]$ and $h \notin (f)$. Two such fractions represent the same rational function precisely when

$$\frac{g}{h} = \frac{g'}{h'} \text{ in } k(C) \iff gh' - g'h \in (f). \quad (5)$$

The denominator must be nonzero *in the coordinate ring*; it need not be nonzero at every point of the curve.

Example 5.2 (The parabola). For $C : y = x^2$,

$$k[C] = k[x, y]/(y - x^2) \simeq k[t],$$

where the classes of x and y correspond to t and t^2 . Consequently,

$$k(C) \simeq k(t).$$

Here $k[t]$ is a polynomial ring, whereas $k(t)$ is its field of rational functions.

Example 5.3 (A quadratic function field). For $C : y^2 = x^3 - x$ in characteristic different from 2,

$$k[C] = k[x, y]/(y^2 - x^3 + x), \quad k(C) = k(x)(y), \quad y^2 = x^3 - x.$$

The notation suppresses the bars on the classes of x and y . The element x is transcendental over k , and $x^3 - x$ is not a square in $k(x)$. Indeed, an identity

$$\left(\frac{a(x)}{b(x)}\right)^2 = x^3 - x$$

with nonzero polynomials a, b would give $2 \deg a = 3 + 2 \deg b$, which is impossible. Therefore

$$[k(C) : k(x)] = 2,$$

and every element of $k(C)$ has a unique expression

$$a(x) + b(x)y, \quad a(x), b(x) \in k(x).$$

This is the quadratic-extension description in Morandi, Example 2.10.¹

Example 5.4 (The singular cubic). For $C : y^2 = x^3$, the coordinate ring is $k[t^2, t^3]$, under $x \mapsto t^2$, $y \mapsto t^3$. Its field of fractions is $k(t)$, since $t = t^3/t^2$. In the notation of the curve, $t = y/x$. Thus the parabola and the cuspidal cubic have isomorphic function fields even though one is smooth and the other is singular. This is Morandi's Example 2.9.

A rational function may be defined at a point even when one chosen expression has a zero denominator there. It is enough that some equivalent expression have a nonzero denominator. Formula (5) determines when such expressions agree.

5.2 The function field of a projective curve

Let $\overline{C} \subseteq \mathbb{P}^2$ be defined by an irreducible homogeneous polynomial $F(X, Y, Z)$. Its homogeneous coordinate ring is

$$S(\overline{C}) = k[X, Y, Z]/(F).$$

It is again an integral domain, but its entire field of fractions is *not* the function field of the projective curve. To obtain scalar-independent values on projective points, we use homogeneous fractions of degree zero.

¹The displayed coordinate ring in Morandi's Example 2.10 has a sign typo: for the equation $y^2 = x^3 - x$, the defining ideal is $(y^2 - x^3 + x)$.

Definition 5.5. The function field $k(\overline{C})$ consists of 0 and equivalence classes of fractions

$$\frac{H(X, Y, Z)}{J(X, Y, Z)},$$

where H, J are homogeneous of the same degree and $J \notin (F)$. The equivalence relation is

$$\frac{H}{J} \sim \frac{H'}{J'} \iff HJ' - H'J \in (F).$$

Addition and multiplication are the usual operations on fractions.

If H and J have common degree m , then

$$\frac{H(\lambda X, \lambda Y, \lambda Z)}{J(\lambda X, \lambda Y, \lambda Z)} = \frac{\lambda^m H(X, Y, Z)}{\lambda^m J(X, Y, Z)} = \frac{H(X, Y, Z)}{J(X, Y, Z)}.$$

This explains both the equal-degree condition and why the quotient gives a well-defined value wherever its denominator does not vanish. Products, sums, and inverses of nonzero homogeneous degree-zero fractions have the same form, so this is a field. This is the definition in Morandi, §2.3, and Walker, Definition 5.8.

5.3 Passing to projective closure does not change the function field

Proposition 5.6 (Morandi, Proposition 2.11). *Let $f(x, y) \in k[x, y]$ be irreducible, let $C : f = 0$, and let $\overline{C} : F = 0$ be its projective closure. Restriction to the affine chart $Z \neq 0$ induces an isomorphism*

$$k(\overline{C}) \simeq k(C).$$

Under this identification, X/Z corresponds to x and Y/Z corresponds to y .

Proof. The homogenization F is irreducible. To see this, a nontrivial homogeneous factorization of F would dehomogenize to a factorization of f . Since f is irreducible, one factor would dehomogenize to a nonzero constant, so that homogeneous factor would be a constant times a power of Z . But Z does not divide F . Hence the factor was already constant.

Define

$$\phi : k[C] \longrightarrow k(\overline{C}), \quad [h(x, y)] \longmapsto h(X/Z, Y/Z).$$

For a nonzero polynomial h of degree r , write h^h for its homogenization. Then

$$h(X/Z, Y/Z) = \frac{h^h(X, Y, Z)}{Z^r},$$

which is a fraction of homogeneous polynomials of the same degree. The class of Z is nonzero because the curve is not the line $Z = 0$.

The map is well defined: if $h' - h = fa$, then

$$h'(X/Z, Y/Z) - h(X/Z, Y/Z) = \frac{F(X, Y, Z)}{Z^{\deg f}} a(X/Z, Y/Z) = 0$$

in $k(\overline{C})$. Substitution also respects addition and multiplication, so ϕ is a ring homomorphism.

To prove injectivity, suppose $\phi([h]) = 0$. If $h \neq 0$, this means that $h^h \in (F)$, say $h^h = FB$. Setting $Z = 1$ yields

$$h(x, y) = f(x, y)B(x, y, 1),$$

so $[h] = 0$ in $k[C]$. Thus ϕ is injective and extends to a field homomorphism

$$\Phi : k(C) \longrightarrow k(\overline{C}).$$

For surjectivity, take a fraction H/J representing an element of $k(\overline{C})$, with H, J homogeneous of the same degree m . By homogeneity,

$$\frac{H(X, Y, Z)}{J(X, Y, Z)} = \frac{H(X/Z, Y/Z, 1)}{J(X/Z, Y/Z, 1)}.$$

The polynomial $J(x, y, 1)$ has nonzero class in $k[C]$: otherwise its image under ϕ , which is J/Z^m , would be zero, contradicting $J \notin (F)$. Therefore

$$\frac{H(x, y, 1)}{J(x, y, 1)} \in k(C)$$

is well defined, and its image under Φ is H/J . This proves surjectivity. $\square$

Remark 5.7. In applying Morandi's proposition, the affine chart must be nonempty. This is automatic for the projective closure of an affine curve. For an arbitrary projective curve contained in $Z = 0$, one must choose a different chart instead. The proof above makes this hypothesis and the substitutions in the homogeneous ratios explicit.

We will henceforth identify these fields and write $k(C) = k(\overline{C})$ when the affine/projective distinction is understood. Passing to projective closure changes the points at which a rational function is examined, but does not change the rational function itself. For example, the affine coordinate x becomes X/Z ; the added points may be poles of that function.

6 Bézout's theorem: counting all intersections

Working projectively and over an algebraic closure leads to a uniform intersection count. The count must include multiplicity.

Theorem 6.1 (Bézout). *Let C and C' be projective plane curves defined over k by homogeneous polynomials of degrees d and e . Assume that they have no common component over $\overline{k}$. Then they meet in finitely many geometric points, and*

$$\sum_{P \in C(\overline{k}) \cap C'(\overline{k})} I_P(C, C') = de,$$

where $I_P(C, C')$ is the intersection multiplicity at P .

We state the theorem without proof, following Walker, Theorem 3.11. The no-common-component hypothesis is essential: otherwise the intersection need not be finite. The theorem counts points over $\overline{k}$, not only over k , and a point with intersection multiplicity m contributes m rather than 1.

For the simple examples below, intersection multiplicities can be read from the multiplicities of roots after restricting the equation of one curve to a line. A general definition is not needed for these computations.

Example 6.2 (Parallel affine lines). The affine lines $y = 0$ and $y = 1$ do not meet. Their projective closures are $Y = 0$ and $Y - Z = 0$. Their intersection is $[1 : 0 : 0]$, a point at infinity, with multiplicity one. This agrees with $1 \cdot 1 = 1$.

Example 6.3 (A parabola and a horizontal line). Intersect $C : YZ = X^2$ with the line $Y = bZ$. The resulting homogeneous equation on the line is

$$X^2 = bZ^2.$$

Assume the characteristic is not 2. If $b \neq 0$, there are two geometric intersection points

$$[\sqrt{b} : b : 1], \quad [-\sqrt{b} : b : 1].$$

They need not be rational over k . For instance, when $k = \mathbb{F}_3$ and $b = 2$, the roots of $X^2 = 2$ lie in $\mathbb{F}_9$ but not in $\mathbb{F}_3$.

If $b = 0$, the equation is $X^2 = 0$. There is just one geometric intersection point, $[0 : 0 : 1]$, but it has multiplicity two. The line is tangent to the parabola there.

Example 6.4 (A parabola and a vertical line). The line $X = aZ$ meets $YZ = X^2$ in

$$[a : a^2 : 1] \quad \text{and} \quad [0 : 1 : 0].$$

Indeed, substitution gives $Z(Y - a^2Z) = 0$. Both intersections have multiplicity one. The second point is the intersection that would be missed by staying in the affine plane.

These examples exhibit the three features needed for the formula *de*: projective points at infinity, points defined over field extensions, and multiplicities.

7 Frobenius and points of higher degree

7.1 Frobenius orbits

Let $k = \mathbb{F}_q$ and $K = \mathbb{F}_{q^n}$. The map

$$\sigma_{q,n} : K \longrightarrow K, \quad a \longmapsto a^q,$$

is an automorphism, called the *Frobenius automorphism* relative to $\mathbb{F}_q$. The power map preserves addition and multiplication, and it is injective, hence bijective because K is finite. It satisfies

$$\sigma_{q,n}^n = \text{id}, \quad \{a \in K : \sigma_{q,n}(a) = a\} = \mathbb{F}_q.$$

The fixed-field statement follows because $T^q - T$ has exactly the q elements of $\mathbb{F}_q$ as its roots.

Let C be a projective plane curve with equation $F = 0$ defined over $\mathbb{F}_q$. Frobenius acts on its K -rational points by

$$\sigma_{q,n}([a : b : c]) = [a^q : b^q : c^q]. \tag{6}$$

This is well defined: replacing (a, b, c) by $(\lambda a, \lambda b, \lambda c)$ replaces its image by a scalar multiple with scalar λ^q . Furthermore,

$$F(a^q, b^q, c^q) = F(a, b, c)^q = 0,$$

because the coefficients of F lie in $\mathbb{F}_q$. Thus Frobenius preserves the curve.

A projective point fixed by Frobenius is rational over $\mathbb{F}_q$. To see this, normalize one nonzero coordinate to 1. The equality of projective points in (6) then says that each normalized coordinate is fixed by $a \mapsto a^q$, hence belongs to $\mathbb{F}_q$.

Definition 7.1 (Points of degree r). A *point of degree r on C over $\mathbb{F}_q$* is an orbit

$$P = \{P_0, \sigma(P_0), \dots, \sigma^{r-1}(P_0)\}$$

of r distinct geometric points under the q -power Frobenius. Equivalently, the P_i lie in $C(\mathbb{F}_{q^r})$ and form an orbit of size exactly r under $\sigma_{q,r}$. We set $\deg P = r$.

This is Walker's convention in Definition 5.4. A degree-one point is an ordinary $\mathbb{F}_q$ -rational point. More generally, a degree- r point is regarded as *one point over the base field*, represented by r conjugate geometric points. These are also called *closed points*.

Remark 7.2. A point in $C(\mathbb{F}_{q^n})$ need not have degree n over $\mathbb{F}_q$. Its Frobenius orbit can have any size r dividing n . The degree is the smallest positive integer r for which the geometric point is rational over $\mathbb{F}_{q^r}$, not merely the degree of an extension in which it happens to have been written.

7.2 A cubic over $\mathbb{F}_3$

For the rest of the lecture, our main example is the projective curve $C_0/\mathbb{F}_3$ whose affine equation is

$$y^2 = x^3 + 2x + 2. \quad (7)$$

This is the example developed in Walker, Chapter 5. In characteristic 3, it is also the equation $y^2 = x^3 - x - 1$ used in Morandi, §6.3.

Its homogeneous equation is

$$Y^2Z = X^3 + 2XZ^2 + 2Z^3.$$

Setting $Z = 0$ gives the unique point at infinity

$$P_\infty = [0 : 1 : 0].$$

The curve is smooth. On the affine chart, the derivative with respect to x of $y^2 - x^3 - 2x - 2$ is 1 in characteristic 3. At infinity, the partial derivative with respect to Z of the homogeneous equation, moved to the left, is 1 at P_∞ .

It is also absolutely irreducible: over any extension of $\mathbb{F}_3$, the cubic $x^3 + 2x + 2$ cannot be a square in the rational function field, by the same odd-degree argument used in Section 5. Thus $y^2 - (x^3 + 2x + 2)$ is irreducible, as is its homogenization.

The rational points over $\mathbb{F}_3$. For every $a \in \mathbb{F}_3$, $a^3 = a$, so

$$a^3 + 2a + 2 = 2.$$

The only squares in $\mathbb{F}_3$ are 0 and 1. Hence there are no affine $\mathbb{F}_3$ -rational points, and

$$\boxed{C_0(\mathbb{F}_3) = \{P_\infty\}}. \quad (8)$$

The points over $\mathbb{F}_9$. The polynomial $T^2 + 1$ is irreducible over $\mathbb{F}_3$, so write

$$\mathbb{F}_9 = \mathbb{F}_3(\alpha), \quad \alpha^2 = -1 = 2.$$

Its elements are $a + b\alpha$ with $a, b \in \mathbb{F}_3$. For $x = 0, 1, 2$, equation (7) becomes $y^2 = 2$, giving $y = \pm\alpha$. To check that these are all the affine points, write

$$x = a + b\alpha, \quad y = c + d\alpha, \quad a, b, c, d \in \mathbb{F}_3.$$

Since $\alpha^3 = -\alpha$, direct computation gives

$$x^3 + 2x + 2 = 2 + b\alpha, \quad y^2 = c^2 + 2d^2 + 2cd\alpha.$$

Thus a solution must satisfy

$$c^2 + 2d^2 = 2, \quad 2cd = b.$$

The first equality forces $c = 0$ and $d = \pm 1$: if $d = 0$ it would require $c^2 = 2$, while if $d \neq 0$ it requires $c^2 = 0$. The second equality then gives $b = 0$. Therefore the complete list is

$$C_0(\mathbb{F}_9) = \{P_\infty\} \cup \{[a : \alpha : 1], [a : -\alpha : 1] : a = 0, 1, 2\}.$$

Frobenius is the cube map, and it sends α to $-\alpha$. Consequently the six affine points form three degree-two points over $\mathbb{F}_3$:

$$\begin{aligned} Q_1 &= \{[0 : \alpha : 1], [0 : -\alpha : 1]\}, \\ Q_2 &= \{[1 : \alpha : 1], [1 : -\alpha : 1]\}, \\ Q_3 &= \{[2 : \alpha : 1], [2 : -\alpha : 1]\}. \end{aligned}$$

In particular,

$$C_0(\mathbb{F}_9) = \{P_\infty\} \sqcup Q_1 \sqcup Q_2 \sqcup Q_3, \quad \#C_0(\mathbb{F}_9) = 1 + 2 + 2 + 2 = 7.$$

There are *three points of degree two over $\mathbb{F}_3$* , not six such points. Over $\mathbb{F}_9$ itself, their six geometric representatives are ordinary rational points.

8 From intersection points to divisors

Suppose C and C' are defined over $\mathbb{F}_q$ and have no common component. Frobenius permutes their geometric intersection points and preserves their intersection multiplicities. We can therefore collect the intersections into Frobenius orbits.

If the distinct orbits are $P_1, \dots, P_t$, with $\deg P_j = r_j$, and the common intersection multiplicity at each geometric point of P_j is m_j , write the *intersection divisor* as

$$C \cap C' = \sum_{j=1}^t m_j P_j. \tag{9}$$

This is a formal sum, not just the set-theoretic intersection. Bézout's theorem becomes

$$\boxed{de = \sum_{j=1}^t m_j \deg P_j = \sum_{j=1}^t m_j r_j.} \tag{10}$$

In Walker's repeated-list notation, one lists P_j a total of m_j times and writes

$$C \cap C' = P_1 + \cdots + P_\ell, \quad \deg = r_1 + \cdots + r_\ell.$$

In that notation, repetitions are permitted. Degree and intersection multiplicity are separate contributions to the count.

Definition 8.1 (Divisors). Let $C/\mathbb{F}_q$ be a smooth projective absolutely irreducible curve. A *divisor on C over $\mathbb{F}_q$* is a finite formal sum

$$D = \sum_P n_P P, \quad n_P \in \mathbb{Z},$$

where P ranges over points of all degrees over $\mathbb{F}_q$, in the sense of Definition 7.1. The divisors form an abelian group $\text{Div}(C)$ under addition of coefficients.

The divisor is *effective*, written $D \geq 0$, if every $n_P \geq 0$. Its support and degree are

$$\text{supp } D = \{P : n_P \neq 0\}, \quad \deg D = \sum_P n_P \deg P.$$

For two divisors, $D \leq E$ means that $E - D$ is effective.

The support is finite by definition. The degree is additive, but equal degree does not imply equal divisors. Moreover, $D \geq 0$ is a condition on *every coefficient*, not merely the inequality $\deg D \geq 0$.

Example 8.2. On $C_0/\mathbb{F}_3$, the divisor $Q_1 - 2P_\infty$ has degree $2 - 2 = 0$, but it is not the zero divisor and is not effective. The single geometric point $[0 : \alpha : 1]$ is not a divisor over $\mathbb{F}_3$ on its own: it must be grouped with its Frobenius conjugate. Their orbit Q_1 is one allowed generator of $\text{Div}(C_0)$, of degree two.

Three line sections of the running cubic. Let L_Z , L_X , and L_Y be the lines $Z = 0$, $X = 0$, and $Y = 0$, respectively. Each intersection with the cubic has total degree three.

The line L_Z meets C_0 only at P_∞ . Therefore, with multiplicity,

$$C_0 \cap L_Z = 3P_\infty. \tag{11}$$

For L_X , setting $X = 0$ gives

$$Z(Y^2 - 2Z^2) = Z(Y - \alpha Z)(Y + \alpha Z) = 0.$$

There are three distinct geometric intersection points. Each has multiplicity one, and over $\mathbb{F}_3$ they group as

$$C_0 \cap L_X = P_\infty + Q_1, \quad 3 = 1 + 2. \tag{12}$$

For L_Y , there is no intersection at infinity, and the affine x -coordinates satisfy

$$h(x) = x^3 + 2x + 2 = 0.$$

This cubic has no root in $\mathbb{F}_3$, hence is irreducible there. Its derivative is 2, so its roots are distinct. Let ω be a root in $\mathbb{F}_{27}$. Then

$$\omega^3 = \omega + 1, \quad \omega^9 = \omega + 2, \quad \omega^{27} = \omega.$$

The three roots form one Frobenius orbit, defining the degree-three point

$$R_1 = \{[\omega : 0 : 1], [\omega + 1 : 0 : 1], [\omega + 2 : 0 : 1]\}.$$

It follows that

$$C_0 \cap L_Y = R_1. \quad (13)$$

Here R_1 has coefficient one and degree three, whereas $3P_\infty$ has coefficient three and degree-one support. Both divisors have degree three, for different reasons.

For a further degree calculation, Walker's example

$$D = 5P_\infty - 2Q_3 + 7R_1$$

has

$$\deg D = 5 \cdot 1 - 2 \cdot 2 + 7 \cdot 3 = 22.$$

9 Divisors of rational functions

9.1 Zeros minus poles

Continue with a smooth projective absolutely irreducible plane curve $C/\mathbb{F}_q$. A nonzero rational function can be written as $\varphi = H/J$, where H and J are homogeneous of the same degree and neither vanishes identically on C .

Definition 9.1 (Principal divisor). The *divisor of φ* is

$$\boxed{\operatorname{div}(\varphi) = (C \cap V(H)) - (C \cap V(J))}, \quad (14)$$

where both intersections are counted with multiplicity and grouped into points over $\mathbb{F}_q$. A divisor of this form is called *principal*.

This is Walker's Definition 5.10. Independence of the chosen fraction is a standard fact about intersection multiplicities on a smooth curve; we use it without proving the general local theory. Common contributions from the numerator and denominator must be canceled.

If

$$\operatorname{div}(\varphi) = \sum_P n_P P,$$

a positive coefficient n_P records a zero of order n_P , and a negative coefficient records a pole of order $-n_P$. Thus

$$\operatorname{div}(\varphi) = \operatorname{div}(\varphi)_0 - \operatorname{div}(\varphi)_\infty,$$

where the zero divisor and pole divisor are effective and have disjoint supports. We also write (φ) for $\operatorname{div}(\varphi)$.

Proposition 9.2. *Every principal divisor has degree zero:*

$$\boxed{\deg \operatorname{div}(\varphi) = 0.}$$

Equivalently, the total degree of the zeros equals the total degree of the poles.

Proof. If C has degree d and H, J have common degree $m > 0$, Bézout gives

$$\deg(C \cap V(H)) = dm = \deg(C \cap V(J)).$$

Subtracting proves the assertion. If $m = 0$, φ is a nonzero constant and its divisor is zero. $\square$

The degree weights and multiplicities are indispensable. This proposition does not say that the number of distinct rational zeros equals the number of distinct rational poles.

Local interpretation and the rules for orders. The coefficient of P in $\text{div}(\varphi)$ is denoted $\text{ord}_P(\varphi)$. On a smooth curve it is the intrinsic order of vanishing. Locally, using a parameter t at a geometric point of P , a nonzero function has the form

$$\varphi = t^r u,$$

where u is regular and nonzero at the point; the integer r is its order. Conjugate points have the same order for a rational function defined over $\mathbb{F}_q$. This local description is the standard smooth-curve fact behind the intersection definition; Morandi's Theorem 2.19 gives its local-ring formulation.

It gives the familiar rules

$$\begin{aligned}\text{ord}_P(\varphi\psi) &= \text{ord}_P(\varphi) + \text{ord}_P(\psi), \\ \text{ord}_P(\varphi/\psi) &= \text{ord}_P(\varphi) - \text{ord}_P(\psi), \\ \text{ord}_P(\varphi + \psi) &\geq \min\{\text{ord}_P(\varphi), \text{ord}_P(\psi)\} \quad \text{if } \varphi + \psi \neq 0.\end{aligned}$$

For the last rule, factor out the smaller power of t . Cancellation may increase the order of the sum, but cannot decrease it. Consequently,

$$\text{div}(\varphi\psi) = \text{div}(\varphi) + \text{div}(\psi), \quad \text{div}(\varphi/\psi) = \text{div}(\varphi) - \text{div}(\psi).$$

There is no analogous additive equality for $\text{div}(\varphi + \psi)$. The zero function is excluded from the definition of a principal divisor.

9.2 Explicit principal divisors

On the running cubic, the affine coordinate functions are the projective rational functions

$$x = X/Z, \quad y = Y/Z.$$

The line sections calculated in (11)–(13) give

$$\text{div}(x) = (P_\infty + Q_1) - 3P_\infty = Q_1 - 2P_\infty, \tag{15}$$

$$\text{div}(y) = R_1 - 3P_\infty. \tag{16}$$

Similarly, the lines $X - Z = 0$ and $X - 2Z = 0$ give

$$\text{div}(x - 1) = Q_2 - 2P_\infty, \quad \text{div}(x - 2) = Q_3 - 2P_\infty.$$

In particular,

$$\text{ord}_{P_\infty}(x) = -2, \quad \text{ord}_{P_\infty}(y) = -3.$$

These pole orders came from intersection multiplicities, not from the apparent $0/0$ expressions at P_∞ .

Example 9.3 (Products and cancellation). For nonnegative integers a, b ,

$$\text{div}(x^a y^b) = aQ_1 + bR_1 - (2a + 3b)P_\infty.$$

The degree is $2a + 3b - (2a + 3b) = 0$, as required. Also,

$$\text{div}(x/y) = Q_1 - R_1 + P_\infty.$$

Thus x/y has a zero, not a pole, at infinity: the pole of the numerator has smaller order than the pole of the denominator.

Example 9.4 (Different expressions for the same function). On C_0 , the relation $y^2 = x^3 - x - 1$ gives

$$x^2 = \frac{y^2 + x + 1}{x}.$$

In homogeneous notation this is

$$\frac{X^2}{Z^2} = \frac{Y^2 + XZ + Z^2}{XZ} \quad \text{in } \mathbb{F}_3(C_0).$$

Both expressions have divisor $2Q_1 - 4P_\infty$. In particular, the zeros of x appearing in the denominator of the second expression do not by themselves give poles: the numerator vanishes there to sufficient order. Walker uses this equality to illustrate independence of the representative in the principal-divisor construction.

For comparison, on the projective line with affine coordinate x , one has

$$\operatorname{div}(x - a) = [a] - \infty, \quad \operatorname{div}\left(\frac{x - a}{x - b}\right) = [a] - [b] \quad (a \neq b),$$

and $\operatorname{div}(x^m) = m[0] - m\infty$ for $m \geq 1$. Near infinity, $u = 1/x$ is a coordinate, so $x^m = u^{-m}$ has a pole of order m .

More generally, a monic irreducible polynomial $h(x) \in \mathbb{F}_q[x]$ of degree r defines a degree- r point P_h on $\mathbb{P}^1$ through its Frobenius orbit of roots. Then

$$\operatorname{div}(h) = P_h - r\infty.$$

For instance, $x^2 + 1$ has no $\mathbb{F}_3$ -rational root, but on $\mathbb{P}_{\mathbb{F}_3}^1$ it has a zero at one point of degree two and a double pole at infinity. Its principal divisor still has degree zero.

10 Riemann–Roch spaces

10.1 Rational functions associated to a divisor

We now reach the function spaces that will replace the low-degree polynomials in Reed–Solomon codes. The definitions and theorem in this section follow Walker, the end of Chapter 5.

Definition 10.1. For a divisor D on a smooth projective absolutely irreducible curve $C/\mathbb{F}_q$, define

$$L(D) = \{\varphi \in \mathbb{F}_q(C)^\times : \operatorname{div}(\varphi) + D \geq 0\} \cup \{0\}.$$

Its dimension over $\mathbb{F}_q$ is denoted $\ell(D)$.

If $D = \sum_P n_P P$, the condition for a nonzero φ is

$$\operatorname{ord}_P(\varphi) \geq -n_P \quad \text{for every } P. \tag{17}$$

When $n_P > 0$, a pole of order at most n_P is allowed. When $n_P < 0$, a zero of order at least $-n_P$ is required. Outside $\operatorname{supp} D$, no poles are allowed.

For example, $L(3P_\infty)$ consists of functions with no poles away from P_∞ and a pole of order at most three there. The space $L(3P_\infty - Q_1)$ additionally requires a zero at each geometric point in the degree-two orbit Q_1 .

Proposition 10.2. *The set $L(D)$ is an $\mathbb{F}_q$ -vector space. Moreover,*

$$D \leq E \implies L(D) \subseteq L(E),$$

$$\deg D < 0 \implies L(D) = \{0\},$$

and

$$\mathbb{F}_q \subseteq L(D) \iff D \geq 0.$$

Proof. Closure under addition is immediate if either summand is zero. For nonzero $\varphi, \psi \in L(D)$ with $\varphi + \psi \neq 0$, the order inequality gives

$$\text{ord}_P(\varphi + \psi) \geq \min\{\text{ord}_P(\varphi), \text{ord}_P(\psi)\} \geq -n_P$$

for every P . If their sum is zero, it belongs to $L(D)$ by definition. Multiplication by a nonzero constant does not change orders; multiplication by zero also stays in $L(D)$. Thus $L(D)$ is a vector space.

If $D \leq E$, the inequalities imposed by E are weaker, proving the inclusion. If a nonzero $\varphi \in L(D)$ existed with $\deg D < 0$, then the effective divisor $\text{div}(\varphi) + D$ would have degree

$$\deg(\text{div}(\varphi) + D) = \deg D < 0,$$

which is impossible. Finally, a nonzero constant has divisor zero, so it belongs to $L(D)$ exactly when D is effective. $\square$

The space $L(D)$ is finite-dimensional. We use this standard fact, as Walker does; its general proof is not needed here. We also use the fact that an everywhere-regular function on a projective absolutely irreducible curve is constant, so

$$L(0) = \mathbb{F}_q.$$

The explicit calculations below exhibit the finite-dimensional spaces directly.

On $\mathbb{P}^1$, for $m \geq 0$,

$$L(m\infty) = \{f(x) \in \mathbb{F}_q[x] : \deg f \leq m\} \cup \{0\} = \langle 1, x, \dots, x^m \rangle_{\mathbb{F}_q}. \quad (18)$$

Indeed, write a rational function as a reduced fraction $a(x)/b(x)$. Any nonconstant denominator has an irreducible factor, which gives a finite pole, possibly at a point of higher degree. Thus a function with no finite poles must be a polynomial. Its pole order at infinity is its degree. Consequently,

$$\ell(m\infty) = m + 1 \quad (m \geq 0).$$

This is exactly the polynomial space used in the Reed–Solomon construction.

10.2 The Riemann–Roch theorem

The *genus* is a nonnegative integer attached to a smooth projective curve. For the smooth projective plane curves considered here, of degree e , it is given by

$$g = \frac{(e-1)(e-2)}{2}.$$

We use this plane-curve genus formula as a geometric input, as in Walker, §4.2. Thus a line or a smooth conic has genus zero, and a smooth plane cubic has genus one. The formula must not be applied to a singular plane equation as a formula for the genus of its smooth model.

Theorem 10.3 (Riemann–Roch, in the form used by Walker). *Let $C/\mathbb{F}_q$ be a smooth projective absolutely irreducible plane curve of genus g , and let D be a divisor on C over $\mathbb{F}_q$. Then*

$$\ell(D) \geq \deg D + 1 - g.$$

If $\deg D > 2g - 2$, then

$$\ell(D) = \deg D + 1 - g.$$

This is the statement of Walker’s Theorem 5.14, and is stated without proof in this lecture. The inequality applies to every divisor; the equality requires the strict degree hypothesis. All dimensions are taken over the original field $\mathbb{F}_q$, not over its algebraic closure.

The theorem explains what changes when the projective line is replaced by a curve of genus g . For a sufficiently large allowed pole divisor, the dimension is its degree plus one, minus g . The degree still controls poles and hence zeros, but the dimension of the allowed space depends on the geometry of the curve as well.

10.3 Riemann–Roch on the running cubic

Our curve $C_0/\mathbb{F}_3$ is a smooth projective cubic, so $g = 1$. For every integer $m \geq 1$, the divisor mP_∞ has degree $m > 2g - 2 = 0$. Therefore

$$\ell(mP_\infty) = m \quad (m \geq 1). \quad (19)$$

At $m = 0$, however, $\ell(0) = 1$. This is an immediate example of why the strict degree hypothesis cannot be omitted.

From (15) and (16), x and y have their only poles at P_∞ , of orders two and three. Their monomials give the following bases:

m	$\ell(mP_\infty)$	Basis of $L(mP_\infty)$ over $\mathbb{F}_3$
0	1	1
1	1	1
2	2	1, x
3	3	1, x , y
4	4	1, x , y , x^2
5	5	1, x , y , x^2 , xy
6	6	1, x , y , x^2 , xy , x^3

Why do these lists give bases? Their elements satisfy the required pole bounds, and their pole orders are distinct. In a nonzero linear combination, the term of largest pole order cannot be canceled by terms of smaller pole order, so the elements are linearly independent. The number of elements equals the Riemann–Roch dimension, proving that they span.

At pole order six, y^2 is not a new independent function: the defining equation says

$$y^2 = x^3 + 2x + 2.$$

More generally, for $m \geq 1$, a basis is

$$\{x^a : a \geq 0, 2a \leq m\} \cup \{x^a y : a \geq 0, 2a + 3 \leq m\}. \quad (20)$$

Its pole orders are precisely $0, 2, 3, \dots, m$, with the list interpreted as just 0 when $m = 1$. There are m elements. This extends Walker's first few calculations at the end of Chapter 5, and agrees with the basis calculation in Morandi, §6.3.

There is no nonconstant function with a single simple pole at P_∞ , since $L(P_\infty) = \mathbb{F}_3$. This is a concrete difference from $\mathbb{P}^1$, where x has a simple pole at infinity and $\ell(\infty) = 2$.

We have now reached the geometric replacement for the polynomial spaces in Lecture 1: a finite-dimensional space $L(D)$, whose allowed poles also bound its zeros. Evaluating such functions at rational points outside $\text{supp } D$ will be the next step toward constructing algebraic geometry codes.

Lecture summary

The main objects and conclusions from Lecture 2 are:

- Shortening preserves the distances among retained codewords while decreasing length. Applied before finite Plotkin, it gives $\alpha_q(\delta) \leq 1 - \delta/(1 - 1/q)$ and zero asymptotic rate at and above the threshold $1 - 1/q$.
- Every finite field has prime-power cardinality. Inside a fixed algebraic closure, $\overline{\mathbb{F}_q} = \bigcup_{n \geq 1} \mathbb{F}_{q^n}$, with the subfields organized by divisibility of extension degrees.
- Homogenization embeds an affine plane curve into a projective plane curve. Smoothness is tested by formal partial derivatives over the algebraic closure, including at infinity.
- The coordinate ring $k[x, y]/(f)$ of an irreducible affine curve is a domain. Its field of fractions agrees with the field of homogeneous degree-zero rational functions on the projective closure.
- Over $\mathbb{F}_q$, a point of degree r is a Frobenius orbit of r geometric points. Bézout's count groups these into an intersection divisor of total degree de , with both orbit degrees and intersection multiplicities included.
- Divisors record zeros and poles. Every principal divisor has degree zero, and $L(D)$ consists of rational functions satisfying the coefficientwise inequality $\text{div}(f) + D \geq 0$. Riemann–Roch gives $\ell(D) \geq \deg D + 1 - g$, with equality for $\deg D > 2g - 2$.

References for this lecture

- J. L. Walker, *Codes and Curves*, especially §2.2 (the asymptotic Plotkin bound), Chapters 3–5 (curves, smoothness, points, divisors, and Riemann–Roch), and Appendix B (finite fields). The numbering refers to the supplied manuscript.
- P. J. Morandi, *Error Correcting Codes and Algebraic Curves*, lecture notes for Mathematics 601, Fall 2001, especially §§2.1–2.5 and Proposition 2.11 (affine and projective function fields), and §6.3 (the cubic over $\mathbb{F}_3$).